

12. Addition of Two Sine Waves Using Vivado HLS

Aim

To design and implement a system for addition of two sine wave signals using FPGA, where the addition circuit is designed using Vivado HLS in C++ and converted into an IP block, and the input sine wave signals are generated using DDS Compiler IP cores. The complete system is verified using a Verilog testbench.

Software Required

- Xilinx Vivado Design Suite
- Vivado HLS
- Vivado IP Catalog
- Vivado Simulator (XSim)
- Verilog HDL

Procedure

- **Adder design**
 - A C++ function was written using fixed-point data type `ap_int<16>` to add two input signals.
 - The function accepts two 16-bit signed inputs and produces a 16-bit signed output.
 - The C++ design was synthesized using Vivado HLS to generate RTL.
- **IP Core Generation**
 - The synthesized HLS design was packaged as a reusable **IP block**.
 - The generated IP was exported to the Vivado IP Catalog.
- **Sine Wave Generation Using DDS Compiler**
 - Two **DDS Compiler IP cores** were configured to generate sine wave signals.
 - Both DDS cores produce fixed-point sine samples compatible with the HLS adder input width.
- **System Integration in Vivado**
 - The two DDS Compiler outputs were connected to the inputs of the HLS-generated adder IP.
 - The adder output represents the summed sine wave signal.
- **Testbench Development**
 - A **Verilog testbench** was written to provide clock and control signals.
 - The testbench monitored the outputs of the DDS blocks and the adder output.
- **Simulation and Verification**
 - The complete design was simulated using Vivado Simulator.
 - The correctness of the summed sine wave was verified by observing waveform outputs.

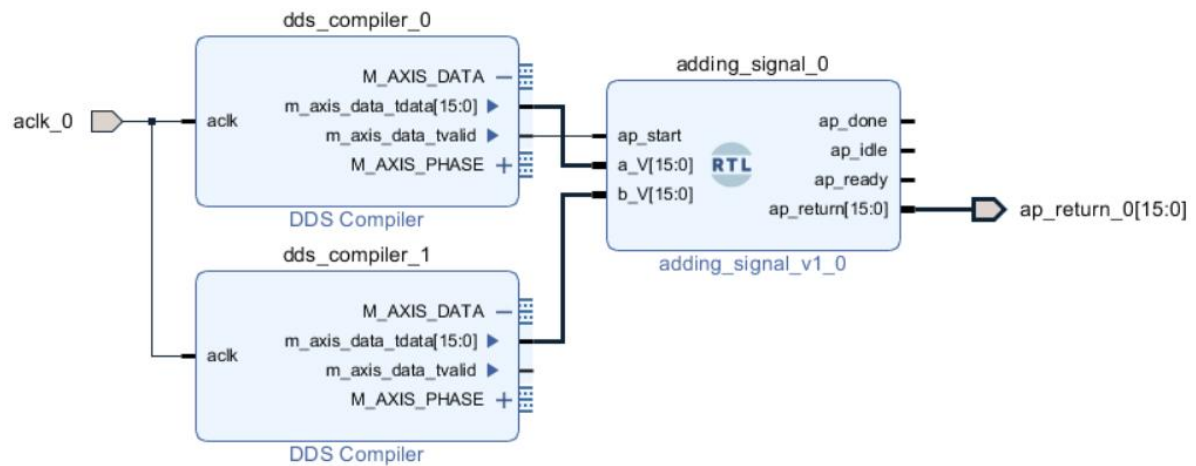
Cpp program

```
#include <ap_int.h>

ap_int<16> adding_signal(ap_int<16> a, ap_int<16> b) {
    return a + b;
}
```

RTL Generated code

```
module adding_signal (
    ap_start,
    ap_done,
    ap_idle,
    ap_ready,
    a_V,
    b_V,
    ap_return
);
input  ap_start;
output ap_done;
output ap_idle;
output ap_ready;
input  [15:0] a_V;
input  [15:0] b_V;
output [15:0] ap_return;
assign ap_done = ap_start;
assign ap_idle = 1'b1;
assign ap_ready = ap_start;
assign ap_return = (b_V + a_V);
endmodule
```



Testbench program

```

module tb_adding_two_signals;

reg clk;

reg rst;

wire clk_out;

design_2_wrapper uut (

    .acclk_0(clk),

    .ap_return_0()

);

always #5 clk = ~clk;

initial begin

    clk = 0;

    rst = 1;

    #20 rst = 0;

    #100;

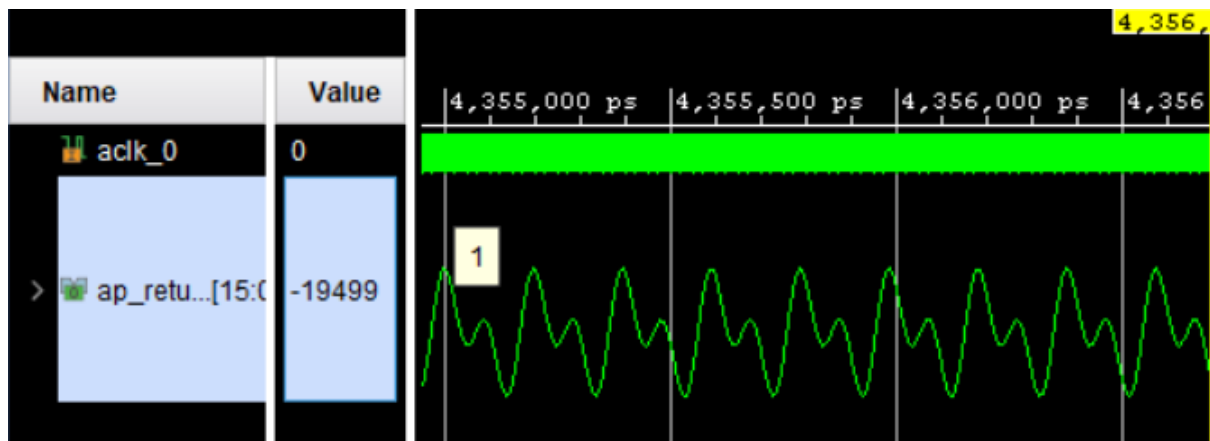
    $stop;

end

endmodule

```

Output



Result

The addition of two sine wave signals was successfully implemented and verified. The sine waves generated using DDS Compiler IP cores were correctly applied to the HLS-designed adder IP block. Simulation results showed that the output waveform corresponds to the arithmetic sum of the two input sine waves, confirming correct functionality of the addition circuit. The design was successfully synthesized, converted into an IP block, and validated using a Verilog testbench.